

BRADFORD, PA, USA

WMO: 725266

Lat: 41.800N

Lon: 78.633W

Elev: 2142

StdP: 13.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04751

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	2.7	-9.9	3.5	-0.4	-5.0	4.5	5.1	22.9	14.4	20.5	17.1	6.3	270	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	83.4	68.7	81.2	67.3	79.0	66.2	71.5	78.7	70.1	77.1	68.7	75.3	7.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	117.2	74.7	67.9	110.8	73.4	66.4	105.1	72.1	36.7	78.8	35.3	76.6	34.1	75.5	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.6	15.9	DB	-11.0	86.9	6.8	3.0	-15.9	89.1	-19.8	90.9	-23.7	92.6	-28.6	94.8	(4)
(5)			WB	-11.5	74.2	6.6	1.9	-16.2	75.6	-20.0	76.7	-23.7	77.8	-28.5	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	22.2	24.5	32.1	44.2	55.0	62.2	65.8	64.6	58.5	47.7	37.3	28.1	(6)	
	DBStd	17.86	12.31	11.08	11.22	9.43	8.56	6.55	5.56	5.34	7.09	8.74	9.34	10.08	(7)	
	HDD50	3649	862	714	566	222	47	2	0	0	9	153	392	682	(8)	
	HDD65	7422	1325	1133	1020	625	327	128	59	74	217	540	830	1144	(9)	
	CDD50	1938	2	1	12	48	201	369	491	453	266	80	12	3	(10)	
	CDD65	233	0	0	0	1	15	44	85	62	23	3	0	0	(11)	
	CDH74	2041	0	0	2	24	198	404	736	464	199	14	0	0	(12)	
	CDH80	316	0	0	0	1	25	60	138	63	29	0	0	0	(13)	
	WSAvg	6.7	8.3	8.2	8.4	7.9	6.4	5.5	4.7	4.5	5.0	6.4	7.2	7.9	(14)	
	PrecAvg	44.4	2.8	2.7	3.3	3.4	4.1	4.6	4.3	3.8	4.1	3.2	3.7	3.7	(15)	
(16) Precipitation	PrecMax	59.1	5.8	5.3	6.3	6.2	8.8	11.2	10.1	6.9	7.7	6.8	8.9	7.3	(16)	
	PrecMin	26.5	0.4	0.7	0.5	1.3	0.9	0.8	1.4	1.7	1.5	0.7	0.4	1.2	(17)	
	PrecStd	7.2	1.4	1.3	1.5	1.3	1.9	2.4	1.7	1.6	1.7	1.3	1.7	1.5	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	58.7	70.3	77.6	83.1	84.5	86.8	85.0	83.7	76.0	65.5	59.4	(19)	
		MCWB	52.2	50.3	56.3	59.0	65.7	68.5	71.1	69.5	69.0	64.1	53.8	52.4	(20)	
	2%	DB	51.8	51.1	61.8	72.1	79.4	81.8	83.7	81.6	79.4	71.1	60.2	53.7	(21)	
		MCWB	49.0	45.0	50.8	55.6	62.7	67.9	69.8	67.9	66.8	60.0	51.5	50.2	(22)	
	5%	DB	45.5	45.8	55.5	66.5	75.4	79.0	80.9	79.1	75.7	67.1	56.5	47.6	(23)	
		MCWB	41.9	40.2	48.5	52.5	61.0	66.3	67.9	67.0	64.8	59.0	50.3	43.5	(24)	
	10%	DB	39.6	40.8	50.0	61.6	71.4	75.5	78.4	76.3	72.2	63.0	53.1	42.3	(25)	
		MCWB	36.3	36.3	43.0	49.6	58.8	64.0	66.6	65.5	63.0	56.7	48.0	39.0	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	52.9	57.5	62.0	68.5	71.4	74.0	73.0	72.2	67.5	57.9	55.4	(27)	
		MCDB	54.8	55.9	68.2	73.0	78.5	79.3	82.9	80.6	79.4	72.5	62.0	58.7	(28)	
	2%	WB	49.1	46.2	53.3	58.4	66.2	69.7	72.0	70.9	69.2	64.0	54.5	50.1	(29)	
		MCDB	51.6	50.0	58.6	66.9	75.0	78.0	80.0	76.9	76.2	68.4	57.8	52.8	(30)	
	5%	WB	42.8	41.2	49.0	55.4	63.9	68.3	70.3	69.3	67.2	60.6	51.2	43.9	(31)	
		MCDB	45.4	44.9	54.9	63.2	71.8	75.7	77.5	75.2	72.7	64.8	55.2	46.9	(32)	
	10%	WB	36.6	36.4	43.7	52.2	61.6	66.8	68.7	67.8	65.1	57.2	47.7	39.1	(33)	
		MCDB	39.5	40.6	49.6	59.8	68.9	73.0	75.1	73.7	69.8	61.9	52.6	42.2	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	14.2	15.9	17.5	21.5	22.4	21.2	21.7	20.9	21.2	18.3	14.9	12.4	(35)	
		MCDBR	17.3	21.2	25.3	30.1	28.2	25.0	24.1	23.8	25.1	24.9	21.1	18.8	(36)	
		MCWBR	15.4	16.3	16.2	16.8	14.0	12.7	11.9	12.0	13.4	14.6	14.9	16.2	(37)	
	5% WB	MCDBR	16.9	20.3	22.4	23.8	22.3	20.5	20.1	19.1	19.1	19.5	17.6	17.8	(38)	
		MCWBR	16.3	17.2	16.4	15.5	12.3	11.4	10.7	10.5	10.9	13.2	14.6	16.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.313	0.327	0.349	0.378	0.418	0.417	0.409	0.354	0.343	0.303	0.286	(40)	
	taud		2.406	2.325	2.398	2.396	2.330	2.238	2.247	2.263	2.409	2.447	2.528	2.475	(41)	
	Ebn at Noon		274	282	292	292	285	272	271	270	279	269	268	266	(42)	
	Edh at Noon		25	32	33	36	40	43	43	41	33	28	22	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		446	695	1032	1390	1625	1765	1800	1611	1296	824	533	352	(44)	
	RadStd		39	83	110	149	166	147	140	96	143	87	76	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+61	(46)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A	(47)

Nomenclature: See separate page